

Solution Requirements

Date	29 June 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Crop Recommendation System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The current system does not require user authentication. Future versions may include secure login for farmers and administrators.	Low
2	Authorization levels	All users can access the crop recommendation service. Future admin access may allow model updates and dataset management.	Low

3	External interfaces	The system interacts with the trained machine learning model and supports REST API requests. Future versions can integrate weather APIs and IoT soil sensors.	High
4	Transactions processing	The system validates user inputs, processes them through the ML pipeline, and returns crop recommendations with prediction probabilities.	High
5	Reporting	Displays recommended crop and alternative crop probabilities through the web dashboard and JSON API response.	Medium
6	Business rules, etc.	Accept only valid N, P, K, temperature, humidity, pH, and rainfall values. Reject invalid inputs before prediction and recommend the crop with the highest probability.	High
7	Compliance to laws or regulations	Follow secure coding practices and protect user input data. No personally identifiable information is collected or stored.	Medium
8	<i>Other</i>	Support both browser-based users and external applications through HTML interface and RESTful API.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	Generate crop recommendations quickly after user submission.	Prediction response in < 1 second
2	Scalability	Support increasing numbers of prediction requests without affecting performance.	Supports multiple concurrent users with consistent response time
3	Security & Data Privacy	Validate all user inputs and prevent invalid or malicious data from entering the prediction pipeline.	100% input validation; no storage of personal user data
4	Reliability & Availability	Ensure consistent prediction results and uninterrupted system operation.	99%+ availability during deployment
5	Usability & Accessibility	Provide an intuitive, responsive interface that works across desktops and mobile devices.	Responsive UI with easy-to-use forms and clear prediction results
6	<i>Other</i>	Modular codebase allowing easy model replacement, updates, and deployment on different platforms.	Modular architecture with reusable components and cross-platform compatibility